HOMOLOGY AS A COLIMIT IN THE SATTLER–WÄRN AXIOMS, AND ITS FORMALISATION IN UNIT

1. Introduction

In homotopy type theory the cohomology of a type X with coefficients in an abelian group A is simple to define:

$$H^n(X; A) = \|X \to K(A, n)\|_0.$$

The function type $X \to K(A,n) = \prod_{x:X} K(A,n)$ is the limit over X of the constant diagram at $K(A,n)$. Limits of groups, spectra or chain complexes are computed on the underlying types, so Π suffices.

Homology is the dual notion: the colimit over X of the constant diagram at the coefficients, taken in a category D of abelian objects (spectra, or chain complexes up to quasi-isomorphism). This colimit is not Σ, because forgetful functors do not preserve colimits. Type theory has native colimits in one category only, the universe. Homology therefore needs colimits indexed by types in a different category. D. Wärn suggested that a theory of higher categories inside type theory could provide them.

This note tests the suggestion in the axiomatic setting of Sattler and Wärn, which we formalised earlier in Unit (`unit/doc/hcat-segal.pdf`). For a constant diagram the colimit is the *copower* $X \otimes A$. We define homology as $X \otimes A$ and prove the Eilenberg–Steenrod axioms at the level of D:

- homotopy invariance;
- dimension, $\mathbf{1} \otimes A \cong A$;
- additivity;
- excision: pushouts of types go to pushouts in D;

as well as a Fubini formula. With a zero object and stability we then define the homology groups, prove the Mayer–Vietoris sequence (in part) and the suspension isomorphism, and compute $\tilde{H}_*(S^n)$ (§7). All of the proofs of the axioms are short consequences of the universal property except excision. Excision needs the pentagon for composition inside D. In this setting that is a coherence of level 3 of Cat (§6).

2. The setting

We work in dependent type theory with univalence, as in `hcat-segal`. Cat is a wild category: a type of categories with types $\mathrm{Map}(C, D)$, composition and identities, and its coherences given as axioms level by level. For a category C:

- $\mathrm{Ob}C = \mathrm{Map}(\mathbf{1}, C)$;
- $\mathrm{Mor}C = \mathrm{Map}(\mathbb{I}, C)$;
- $\mathrm{Hom}_C(x, y)$ consists of the morphisms with endpoint paths to x and y.

Composition gf of $g : \mathrm{Hom}(x, y)$ and $f : \mathrm{Hom}(y, z)$ (written in diagrammatic order) is the long edge of the centre of the contractible type of 2-simplices with spine (g, f); this uses the Segal axiom. Associativity

$$\alpha(f, g, h) : (fg)h = f(gh)$$

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comes from Segal at $n = 3$ and uses the pentagon of Cat. The unit laws are similar. Every category is univalent (the Rezk axiom). In particular, a functor from a groupoid X into D is a function $X \to \mathrm{Ob}D$.

3. Copowers

Definition 3.1. Let D be a category. A *copower* of $A : \mathrm{Ob}D$ by a type X is an object $X \otimes A$ with maps $\eta_x : A \to X \otimes A$ for $x : X$, such that for every T

$$\mathrm{restrict} : \mathrm{Hom}(X \otimes A, T) \to (X \to \mathrm{Hom}(A, T)), \qquad g \mapsto (x \mapsto \eta_x g)$$

is an equivalence. We write lift for its inverse.

$X \otimes A$ is the colimit of the constant X-diagram at A, and we read it as the homology of X with coefficients A. In the intended model D is the category of spectra and $X \otimes A = \Sigma^\infty X_+ \wedge A$, so that $\pi_n(X \otimes HZ) = H_n(X; Z)$.

Remark 3.2 (Size). The objects of a category form a type in a universe U_l. We assume copowers only by types in a strictly smaller universe U_k, $k < l$. With copowers by all types of U_l, Freyd's argument would make D a preorder: take $X = \Sigma_C \mathrm{Hom}(C, T)$ and compare the sizes of $\mathrm{Hom}(X \otimes A, T)$ and $X \to \mathrm{Hom}(A, T)$.

Two consequences of the universal property are used everywhere.

- *β-law*: $\eta_x \,\mathrm{lift}(h) = h(x)$.
- *Extensionality*: two maps out of $X \otimes A$ that agree after every η_x are equal, since restrict is injective and there is function extensionality.

4. The axioms without 2-cells

For $f : X \to Y$ let $\mathrm{t}f = \mathrm{lift}(x \mapsto \eta_{f(x)}) : X \otimes A \to Y \otimes A$. Then $\eta_x \,\mathrm{t}f = \eta_{f(x)}$, and by associativity

$$\eta_x(\mathrm{t}f\, g) = \eta_{f(x)}\, g. \tag{4.1}$$

Proposition 4.1 (Functoriality)**.** $\mathrm{t}(\mathrm{id}) = \mathrm{id}$, $\mathrm{t}(g \circ f) = \mathrm{t}f\,\mathrm{t}g$, *and homotopic maps have equal images.*

Proof. By extensionality. After η_x both sides of each equation become η_x, $\eta_{g(f(x))}$, and $\eta_{f(x)} = \eta_{f'(x)}$ respectively, by the β-law and (4.1). □

Theorem 4.2.

(1) *(Homotopy invariance) If $f : X \to Y$ is an equivalence, then $\mathrm{t}f$ is invertible.*
(2) *(Dimension) $\eta_\star : A \to \mathbf{1} \otimes A$ is invertible.*
(3) *$0 \otimes A$ is initial.*
(4) *(Additivity) $(X + Y) \otimes A$, with $\mathrm{t}\,\mathrm{inl}$ and $\mathrm{t}\,\mathrm{inr}$, is a coproduct of $X \otimes A$ and $Y \otimes A$.*
(5) *(Fubini) $(X \times Y) \otimes A \cong X \otimes (Y \otimes A)$.*

Proof. (1) The inverse is $\mathrm{t}f^{-1}$, by functoriality.

(2) The inverse is $r = \mathrm{lift}(_ \mapsto \mathrm{id}_A)$. We have $\eta_\star r = \mathrm{id}$ by the β-law. For $r\eta_\star = \mathrm{id}$ use extensionality: $\eta_\star(r\eta_\star) = (\eta_\star r)\eta_\star = \eta_\star$.

(3) $\mathrm{Hom}(0 \otimes A, T) \simeq (0 \to \mathrm{Hom}(A, T))$ is contractible.

(4) The inverse of $g \mapsto (\mathrm{t}\,\mathrm{inl}\, g, \mathrm{t}\,\mathrm{inr}\, g)$ sends (h_1, h_2) to the lift of the family $\mathrm{inl}\, x \mapsto \eta_x h_1$, $\mathrm{inr}\, y \mapsto \eta_y h_2$. Both composites are identities by extensionality and (4.1).

(5) The map $(X \times Y) \otimes A \to X \otimes (Y \otimes A)$ is the lift of $(x, y) \mapsto \eta_y \eta_x$. Its inverse is the lift of $x \mapsto \mathrm{lift}(y \mapsto \eta_{(x,y)})$. Both composites are identities by extensionality, applied twice for the second one, together with associativity and the β-law. □

Each proof uses only associativity and the unit laws of D. So all of them hold at level 2 of Cat.

5. Excision

There are no higher inductive types here, so a pushout of types is characterised by its universal property.

Definition 5.1. A commuting square of types

$$\begin{array}{ccc} C & \xrightarrow{\;i\;} & B \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle u} \\ E & \xrightarrow[\;v\;]{} & X \end{array} \qquad h(c) : u(i(c)) = v(j(c))$$

is a *pushout* if for every type T (in the universe of the hom types of D) the map $f \mapsto (f \circ u, f \circ v, \mathrm{ap}_f \circ h)$ from $X \to T$ to the type $\mathrm{Cocone}(T)$ of triples (b, e, θ), with $\theta(c) : b(i(c)) = e(j(c))$, is an equivalence.

A square $i'u' = j'v'$ in D, with 2-cell s, is a *pushout* if for every T the map

$$g \mapsto \left(u'g,\ v'g,\ \alpha^{-1} \cdot (s\, g) \cdot \alpha\right) : \mathrm{Hom}(P, T) \to \mathrm{Cocone}_D(T)$$

is an equivalence. Here $\mathrm{Cocone}_D(T)$ consists of triples (b, e, p) with $p : i'b = j'e$, and the path part is $i'(u'g) = (i'u')g = (j'v')g = j'(v'g)$.

Theorem 5.2 (Excision)**.** *Assume the pentagon for composition in D. Then a pushout square of types is sent to a pushout square in D: $\mathrm{t}i, \mathrm{t}j, \mathrm{t}u, \mathrm{t}v$ with the 2-cell σ defined by $\eta_c\, \sigma =$ the path $\eta_c(\mathrm{t}i\, \mathrm{t}u) = \eta_{u(i(c))} = \eta_{v(j(c))} = \eta_c(\mathrm{t}j\, \mathrm{t}v)$ built from (4.1) and $h(c)$.*

Proof. Let $T : \mathrm{Ob}D$ and $H = \mathrm{Hom}(A, T)$. Define $\psi : \mathrm{Cocone}_D(T) \to \mathrm{Cocone}(H)$ by

$$(b, e, p) \mapsto (\mathrm{restrict}\, b,\ \mathrm{restrict}\, e,\ \theta_p), \qquad \theta_p(c) = \eta_c\, p \text{ conjugated by (4.1) at } i \text{ and } j.$$

ψ is an equivalence: restrict is an equivalence on both legs, and on paths $p \mapsto (c \mapsto \eta_c\, p)$ is an equivalence, being ap of restrict followed by function extensionality.

The square of maps

$$\begin{array}{ccc} \mathrm{Hom}(X \otimes A, T) & \xrightarrow{\mathrm{comparison}_D} & \mathrm{Cocone}_D(T) \\ {\scriptstyle \mathrm{restrict}}\downarrow & & \downarrow{\scriptstyle \psi} \\ (X \to H) & \xrightarrow[\mathrm{comparison}]{} & \mathrm{Cocone}(H) \end{array}$$

commutes up to homotopy. On the legs this is (4.1). On the path part we compare, for each c, two paths in H. Both are composites of:

- the associators at $(\eta_c, \mathrm{t}i, \mathrm{t}u, g)$ and at $(\eta_c, \mathrm{t}j, \mathrm{t}v, g)$;
- instances of (4.1) whiskered by g;
- the path $\mathrm{ap}_\eta(h(c))$.

Each instance of (4.1) contains an associator, and composing it with the associator of the comparison map rebrackets the four-fold composite $\eta_c\, (\mathrm{t}i)\, (\mathrm{t}u)\, g$. Two such rebracketings agree exactly by the pentagon at $(\eta_c, \mathrm{t}i, \mathrm{t}u, g)$, and similarly at j. After that the equation is a groupoid identity.

The left and bottom maps are equivalences, by the copower and by the pushout of types at H. Hence the top map is one. □

6. The coherence needed

The pentagon for composition inside a category C is not an axiom: composition and α are defined from contractible spaces of simplices. To prove it from the Segal condition at $n = 4$ one compares the five associators of a 4-simplex T through its faces. That requires relating

- composites of the form $T \circ S \circ s$, where S is a face and s a 2-simplex of it, with

- the action of T on the composites of S.

This is a coherence for composition in Cat one level up: *internal* level 2 (the pentagon of C) needs *external* level 3. It is the same shift as one level lower, where associativity in C needs the pentagon of Cat. It is also the effect described by Sattler: an n-coherent copy of Cat inside Cat needs $(n+1)$-coherence.

The level 3 coherence of Cat is Stasheff's associahedron K_5. Take five maps d, e, h, g, f and write $a(x, y, z) : (xy)z = x(yz)$. There are two extreme paths from $(((fg)h)e)d$ to $f(g(h(ed)))$:

- L, the three-step path $a((fg)h, e, d) \cdot a(fg, h, ed) \cdot a(f, g, h(ed))$;
- R, the six-step path through $((f(gh))e)d$.

There are two ways to paste the faces of K_5 into a 2-path $L = R$:

- route 1 uses the pentagons at (f, g, h, ed), (f, gh, e, d) and (g, h, e, d) (the last whiskered by f), and the naturality squares of $a(-, e, d)$ and $a(f, -, d)$;
- route 2 uses the pentagons at (fg, h, e, d), (f, g, he, d) and (f, g, h, e) (the last whiskered by d), and the naturality square of $a(f, g, -)$.

The axiom says the two routes agree.

In the proof of the internal pentagon, K5 is used where the action of functors on morphisms (with their endpoints) is compared for three functors at once. At $(F, G, K, m, \mathbb{I}_0)$, the endpoint 2-cells of the two sides follow route 1 and route 2.

If composition in Cat were strictly associative and unital, the pentagons and K_5 would be trivial, and the internal pentagon would follow from Segal at $n = 4$ by routine path algebra. But it would no longer be visible which coherence the argument uses.

7. Homology groups

Stability. Assume a zero object 0 (both $\mathrm{Hom}(0, T)$ and $\mathrm{Hom}(B, 0)$ contractible), write 0 also for the zero maps, and assume suspensions: for each B a pushout square $0 \leftarrow B \rightarrow 0$ with apex ΣB and 2-cell m_B. For *stability* we take the representable form of "pushout squares are pullbacks":

Axiom 7.1 (Stability)**.** For every pushout square $iu = jv$ in D (with $i : C \to B$, $j : C \to E$) and every S, the map $c \mapsto (ci,\ cj,\ \alpha \cdot c\,s \cdot \alpha^{-1})$ from $\mathrm{Hom}(S, C)$ to the type of triples (b, e, p), $p : bu = ev$, is an equivalence.

It needs no pullbacks in D, and it is exactly what the arguments below use. Of "Σ is an equivalence" it gives full faithfulness; essential surjectivity (desuspensions) is used only for $\tilde{H}_k(S^n)$ with $k \leq n$.

Lemma 7.2 (Loops)**.** *$L : \mathrm{Hom}(\Sigma S, T) \simeq \Omega(\mathrm{Hom}(S, T), 0)$, natural in T: for $f : T \to T'$, $L(gf)$ is $L(g)f$ conjugated by the path $0f = 0$, and $L(0) = \mathrm{refl}$. With stability also $\sigma : \mathrm{Hom}(B, E) \simeq \Omega(\mathrm{Hom}(B, \Sigma E), 0)$.*

Proof. $\mathrm{Hom}(\Sigma S, T)$ is the type of cocones (b, e, p) with $b, e : 0 \to T$; as $\mathrm{Hom}(0, T)$ is contractible this is the type of loops at 0. Naturality: both sides are conjugates of $m_S(gf)$, and the two conjugating paths agree by the pentagon at $(0, 0, g, f)$ and naturality of α; this is the second use of the pentagon of D. $L(0) = \mathrm{refl}$ since $0 : \Sigma S \to T$ factors through 0, and $\mathrm{Hom}(S, 0)$ is contractible. σ is stability for $0 \leftarrow E \rightarrow 0$. □

Definition 7.3. Let $\mathbf{1} : \mathrm{Ob}D$ and $S^n_D = \Sigma^n \mathbf{1}$. Then $H_n(X; A) = \pi_0 \mathrm{Hom}(S^n_D, X \otimes A)$, and reduced homology is $\tilde{H}_n(X; A) = \pi_0 \,\mathrm{Red}(S^n_D, X)$, where $\mathrm{Red}(S, X)$ is the fibre over 0 of $c \mapsto c\,\mathrm{t}(!)$, $\mathrm{Hom}(S, X \otimes A) \to \mathrm{Hom}(S, \mathbf{1} \otimes A)$.

$\mathrm{Red}(S, X)$ is $\mathrm{Hom}(S, -)$ of the fibre of $X \otimes A \to \mathbf{1} \otimes A$, so no cofibres in D are needed.

Theorem 7.4 (Groups)**.** *$H_{n+1}(X;A)$ is a group, through $\pi_0 L$ from $\pi_1(\mathrm{Hom}(S^n_D, X \otimes A), 0)$, and it is abelian for $n \geq 1$ (Eckmann–Hilton through L). With stability $H_0(X;A)$ is a group (through σ) and all these groups are abelian.*

Mayer–Vietoris.

Theorem 7.5. *Let $iu = jv$ be a pushout square in D and $S : \mathrm{Ob}D$; write i_* etc. for postcomposition on $\pi_0\mathrm{Hom}(S, -)$.*

1. *$u_*i_* = v_*j_*$, and if $u_*\beta = v_*\varepsilon$ then merely $\beta = i_*\gamma$ and $\varepsilon = j_*\gamma$ for some γ.*
2. *There is $\delta : \pi_0\mathrm{Hom}(\Sigma S, P) \to \pi_0\mathrm{Hom}(S, C)$ with $i_*\delta = 0$, $j_*\delta = 0$, and every γ with $i_*\gamma = 0 = j_*\gamma$ is merely δ of an element.*

Proof. By stability $\mathrm{Hom}(S, C)$ is the type of triples $(b, e, p : bu = ev)$. (1): a path $bu = ev$ gives the triple, hence c. (2): δg corresponds to the triple $(0, 0, p_g)$ where p_g is $L(g)$ conjugated by $0u = 0$ and $0v = 0$; conversely paths $ci = 0$, $cj = 0$ move the triple of c to one of this form. □

With excision this gives, for a pushout of types, $H_{n+1}(X) \xrightarrow{\delta} H_n(C) \to H_n(B) \times H_n(E) \to H_n(X)$, exact at $H_n(C)$ and in the middle.

Theorem 7.6 (Positive-degree exact sequence)**.** *For every $r \geq 2$, a single structural connecting homomorphism gives*

$$H_r(B) \times H_r(E) \xrightarrow{u_* - v_*} H_r(X) \xrightarrow{\partial} H_{r-1}(C) \xrightarrow{(i_*, j_*)} H_{r-1}(B) \times H_{r-1}(E) \xrightarrow{u_* - v_*} H_{r-1}(X).$$

Every arrow is a homomorphism of the specified abelian groups and the sequence is exact at its three interior terms. Consecutive pieces agree.

Proof. For any pointed map, the equivalence between loops of its fibre and the fibre of its loop map retains the projection. Taking components therefore gives both homomorphisms and exactness for the same maps. Apply this first to the projection of the representable pullback onto the product of its legs, whose fibre is the loop space of the apex. Apply it again to the boundary: its fibre is the loop space of that product, over the difference of loop maps. Looping difference gives ordinary group difference on π_2.

The double-loop equivalence obtained from L identifies these with the existing homology groups and maps. Its naturality uses L's already proved naturality and one general lemma: homotopic pointed maps into a type with commuting loops induce equal loop maps, regardless of their chosen pointing paths. Conjugation by a loop is trivial, so no new compatibility of those paths is needed. Exactness at the product is the preceding pullback argument and group cancellation. Difference is additive because the target is abelian. □

This is the positive tail ending at $H_1(X)$, with no exactness claim at that final term. Its extension through degree zero and categorical naturality remain open. The structural boundary need not be compared with the earlier direct formula to obtain these results.

Suspension and spheres. A *suspension* of a type X is a pushout ΣX of $\mathbf{1} \leftarrow X \to \mathbf{1}$, with points N, S; it is given by its universal property.

Theorem 7.7 (Suspension)**.** $\mathrm{Red}(\Sigma S, \Sigma X) \simeq \mathrm{Red}(S, X)$*; hence* $\tilde{H}_{n+1}(\Sigma X; A) \cong \tilde{H}_n(X; A)$.

Proof. Let $P = \Sigma X \otimes A$, $t_N, t_S : \mathbf{1} \otimes A \to P$ and $r : P \to \mathbf{1} \otimes A$ the maps of N, S and $\Sigma X \to \mathbf{1}$, and $F = \mathrm{Red}(S, \Sigma X)$, the fibre of r_*, pointed at 0. We show that both sides are ΩF.

By excision and stability, $\mathrm{Hom}(S, X \otimes A)$ is the type of triples $(a, b, p : at_N = bt_S)$, the first leg being $c \mapsto c\,\mathrm{t}(!)$. So $\mathrm{Red}(S, X)$ is the type of pairs $(b,\ 0t_N = bt_S)$, i.e. the fibre of $(t_S)_*$ over 0. Since $t_S r = \mathrm{t}(\mathbf{1} \to \mathbf{1}) = \mathrm{id}$, $(t_S)_*$ has the retraction r_*. For any such pair of maps the fibre of s

over $s(y_0)$ is Ω of the fibre of r (the total space of the composite fibration is contractible). Moving base points along zero paths gives ΩF.

On the other side, change the base of $\mathrm{Red}(\Sigma S, \Sigma X)$ along L: $gr = 0$ iff $L(gr) = L(0)$, i.e. by Lemma 7.2 iff $L(g)\,r$ conjugated is refl. So it is the fibre over refl of Ωr_*, which is ΩF. □

Let $S^0 = \Sigma\emptyset$ and $S^{n+1} = \Sigma S^n$. Since $\emptyset \otimes A$ is initial, $\mathrm{Hom}(S, \emptyset \otimes A)$ is contractible, so $\mathrm{Red}(S, \emptyset) \simeq \Omega\mathrm{Hom}(S, \mathbf{1} \otimes A) \simeq \mathrm{Hom}(\Sigma S, \mathbf{1} \otimes A)$.

Corollary 7.8 (Spheres)**.** $\tilde{H}_{n+j+1}(S^n; A) \cong \pi_0\mathrm{Hom}(S^{j+1}_D, A) = H_{j+1}(\mathbf{1}; A)$. *With desuspensions* ($\Sigma\Sigma^{-1}B = B$)*,* $\tilde{H}_{n+j}(S^n; A) \cong \pi_0\mathrm{Hom}(S^j_D, A)$ *for all* j*, in particular* $\tilde{H}_n(S^n; A) \cong \pi_0\mathrm{Hom}(\mathbf{1}, A) = H_0(\mathbf{1}; A)$*, and* $\tilde{H}_k(S^{k+m}; A) \cong \pi_0\mathrm{Hom}(\Sigma^{-m}\mathbf{1}, A)$.

Proof. The suspension theorem n times, then $\emptyset$, and the dimension axiom $\mathbf{1} \otimes A \cong A$. With desuspensions write $B = \Sigma\Sigma^{-1}B$ first. □

These are bijections of sets; compatibility of the suspension and sphere isomorphisms with the specified group structures remains unproved.

8. Formalisation

The development is in `unit/sag/hcat`, continuing the port of `hcat-segal`. Segments are nested by level: `Theory` ⊃ `Wild` ⊃ `Structure` ⊃ `Coh2`. Inside `Coh2` there are two new segments.

- `Coh3`, with parameter `k5 : route1 = route2`; the routes are defined in `CatK5`.
- `Homology`, with parameters: a level $k < l$, the category `D`, `tensor`, `eta`, and `tensor-up` (restrict is an equivalence). Inside it, the sub-segment `WithPentagon` has the parameter `hpent`, the pentagon of D (stated in `HcompPentagon`). Inside that, `Stable` has the zero object, suspensions, `stable` and **1**; inside it `WithSpheres` has the suspensions of types and the desuspensions.

Result	File, name
lift, β-law, extensionality	`Copower`: `lift`, `lift-beta`, `ext`
functoriality, (4.1)	`TensorMap`: `tmap-id`, `tmap-comp`, `restrict-tmap`
homotopy invariance, dimension, initial	`TensorInvariance`
additivity	`TensorSum`: `SumOf.tensor-sum`
Fubini	`Fubini`: `FubiniOf.fubini`
pushouts of types, homotopy induction	`TypePushout`
pushouts in D	`DPushout`
the pentagon step	`ExcisionAssoc`: `rt-assoc`
excision	`Excision`: `ExcisionOf.excision`
K_5	`CatK5`, parameter `Coh3.k5`
zero maps, cones	`Zero`, `Cone`
Lemma 7.2	`Loops`: `L`, `L-nat`, `L-zero`; `StableLoop`
groups	`HGroups`: `hgrp`, `hgrp-ab`, `hgrp0`, `hgrp-ab-stable`
Mayer–Vietoris	`MayerVietoris`: `MV`, `MVTypes`
positive-degree exact sequence	`HomologySequence`: `Types`
suspension	`Suspension`: `SuspOf.susp-red`, `susp-iso`
spheres	`Spheres`: `sphere-above`, `sphere-homology`

The pentagon of D (`hpent`) supplies the lemma `rt-assoc` of excision, `L-nat`, and the compatibility of zero-postcomposition paths in `ZeroCoherence`. The type theory of §7 (sets of components, loop

groups, Eckmann–Hilton, fibres) is in `Pi0`, `EckmannHilton`, `FibreAlg`, `Conj`, partly linked from `sag/em`, with the further fibre and difference lemmas described above. The tree has 169 modules and 3978 declarations; a fresh full check with unfolding diagnostics takes 17.44 s with no failures. Each new file checks in under 1 s. The homology sequence does not depend on `Coh3`.

There are no new axioms beyond:

- the parameters `univ`, `images` and `surjections` of `Theory`;
- the axioms of Cat in `Wild`, `Structure`, `Coh2` and `Coh3`;
- the parameters of `Homology`;
- `hpent`;
- the parameters of `Stable`: `zobj`, `zero-init`, `zero-term`, `susp`, `merid`, `susp-po`, `stable`, `unitD`;
- the parameters of `WithSpheres`: `tsusp`, `tnorth`, `tsouth`, `tmerid`, `tsusp-po`, `desusp`, `desusp-eq`.

The formal proofs include the path, fibre and quotient lemmas used in these arguments. The positive exact sequence is assembled from those reusable lemmas in `HomologySequence`; it adds no axioms.

Open.

- Deriving `hpent` from `k5`. The plan is in `sag/hcat/DESIGN-HOMOLOGY.md`; one lemma of the needed calculus of paths between morphisms with endpoints (`HomEqCalc`) is done.
- Degree-zero homomorphisms and the extension of the structural exact sequence through degree zero; categorical naturality of that sequence; the group form of the suspension isomorphism.